GRADE 12 / Physical Sciences / English

Momentum and collisions

01 Understand

Learning goals: Calculate signed momentum • Apply conservation to a closed system • Relate impulse to force and time

Momentum $p = mv$ is a vector. Select a positive direction and give opposite velocities opposite signs.

The total momentum of an isolated system is conserved. In a collision, use the condition of negligible net external impulse.

Impulse equals change in momentum: $F_{net}\Delta t = \Delta p$. Momentum conservation does not imply kinetic energy conservation in an inelastic collision.

Two trolleys stick together; right is positive

Object	Mass	Velocity before	Momentum before
A	2 kg	+3 m·s^{-1}	+6 kg·m·s^{-1}
B	1 kg	0	0
Together	3 kg	+2 m·s^{-1} after	+6 kg·m·s^{-1} after

02 Follow the reasoning

A 2 kg trolley moving right at 3 m·s^{-1} sticks to a stationary 1 kg trolley. Find their final velocity.

1. Before: total momentum = 2(3) + 1(0) = 6 kg·m·s^{-1}.

2. After: the combined mass is 3 kg, so $6 = 3v$ and $v = 2$ m·s^{-1} right.

3. Initial kinetic energy is 9 J; final kinetic energy is 6 J. The 3 J difference becomes other energy forms.

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Practise with purpose

03 Try before checking

Use paper or your notebook. Show your reasoning. For written responses, use evidence and explain your choices. Suggested time: 15–20 minutes.

1. Find the momentum of a 0.5 kg ball moving left at 4 $m \cdot s^{-1}$, taking right as positive.

2. It rebounds right at 6 $m \cdot s^{-1}$. Find Δp.

3. If contact lasts 0.1 s, find the average net force.

04 Check and correct

1. -2 $kg \cdot m \cdot s^{-1}$.

2. Final momentum = +3; change = $3 - (-2) = +5$ $kg \cdot m \cdot s^{-1}$.

3. Fnet = 5/0.1 = 50 N right.

Common mistake to avoid

Change means final minus initial. Reversing direction requires a negative initial or final velocity.

Reflect and revisit

Identify one correction in your own words. Attempt a similar question tomorrow without looking at the guide. If you are still unsure, take your attempt to a teacher or tutor.

Coverage and sources

CAPS topic focus: Mechanics: momentum and impulse

DBE CAPS, Grades 10–12: Physical Sciences. Section 3 content and skills. This is a focused supplement, not a complete syllabus or prescribed literature replacement.

Open the official DBE curriculum reference

Content and visuals: original SmartBridge FET. AI-assisted authoring; subject-teacher and language-specialist review pending. Report an issue with the pack ID phsc-12.